

IAS PRELIMS/MAINS

Ecology & Environment

A stylized illustration of a green city skyline with various skyscrapers, wind turbines, and a sun, set against a light green background with a globe-like shape.

A COMPREHENSIVE COVERAGE

MODULE - 21

AIR POLLUTION

INDOOR AIR POLLUTION

- Indoor air pollution is the degradation of indoor air quality.
- It can be up to 10 times worse than outdoor air pollution. This is because contained areas cause pollutants to build up more than open spaces.
- In developing countries, health impacts of indoor air pollution far outweigh those of outdoor air pollution.

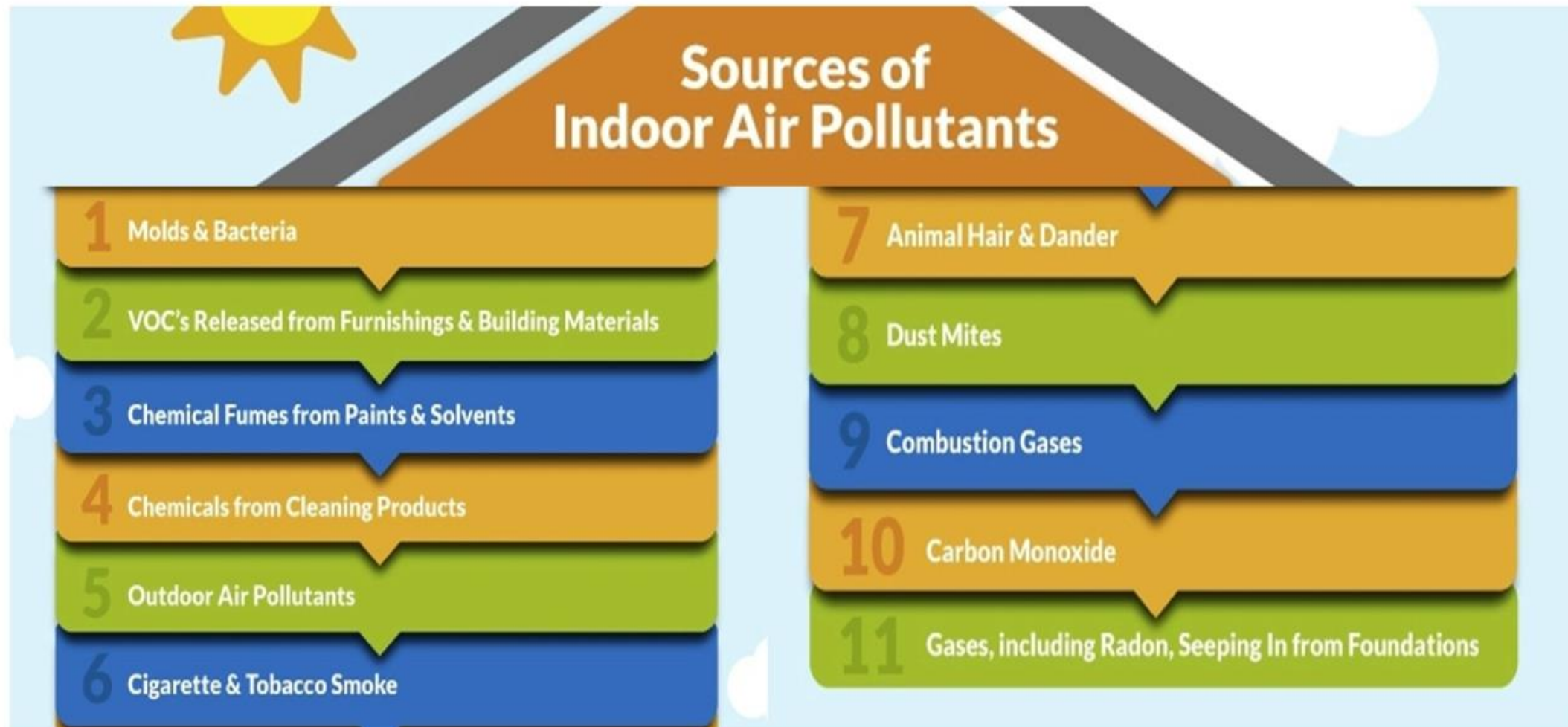


COMBUSTION

- The incomplete combustion products of biomass fuels include suspended particulate matter, carbon monoxide, aromatic hydrocarbons, formaldehyde etc., which have adverse effects on health.
- The combustion of coal results in production of oxides of sulfur, arsenic particles etc.
- Pollutants such as Volatile Organic Compounds are produced from paints, waxes, polishing materials, cosmetics etc.
- Biological pollutants like dust mites, molds, pollen, and infectious agents produced in stagnant water, mattresses and carpets.
- Building material, radon, asbestos, heavy metal particles, volatile organic matter, and tobacco smoke are considered major indoor pollutants in developed countries.
- The combustion products of biomass fuels contribute most to indoor air pollution in developing nations.



COMBUSTION



GOVERNMENT INITIATIVES TO COMBAT AIR POLLUTION

NATIONAL AIR QUALITY MONITORING PROGRAMME

Central Pollution Control Board (CPCB) has been executing a nationwide programme of ambient air quality monitoring known as National Air Quality Monitoring Programme (NAMP).

- The National Air Quality Monitoring Programme (NAMP) is undertaken:
 - ✓ to determine status and trends of ambient air quality.
 - ✓ to ascertain the compliance of National Ambient Air Quality Standards (NAAQS).
 - ✓ to identify non-compliant cities and towns. And,
 - ✓ to undertake preventive and corrective measures.
- The Central Pollution Control Board (CPCB) of India is a statutory body under the Ministry of Environment and Forests (MoEF).
- It was established in 1974 under the Water (Prevention and Control of pollution) Act, 1974.
- CPCB is also entrusted with the powers and functions under the Air (Prevention and Control of Pollution) Act, 1981.



GOVERNMENT INITIATIVES TO COMBAT AIR POLLUTION

NATIONAL AMBIENT AIR QUALITY STANDARDS (NAAQS)

- National Ambient Air Quality Standards are the standards for ambient air quality.
- They are set by the Central Pollution Control Board (CPCB). They are applicable nationwide. The CPCB has been conferred this power by the Air (Prevention and Control of Pollution) Act, 1981.
- The NAAQS are set for 12 pollutants, which include

- ✓ Sulphur dioxide (SO_2)
- ✓ Nitrogen dioxide (NO_2)
- ✓ Particulate matter having micron (PM₁₀)
- ✓ Particulate matter having size less than 2.5 micron (PM_{2.5})
- ✓ Ozone

- ✓ Lead
- ✓ Carbon monoxide (CO)
- ✓ Arsenic
- ✓ Nickel
- ✓ Benzene
- ✓ Ammonia and
- ✓ Benzopyrene.

NAAQS are the maximum permissible limits of pollutants in the ambient air.

NAAQS ARE THE MAXIMUM PERMISSIBLE LIMITS OF POLLUTANTS IN THE AMBIENT AIR.



Pollutant	Time weighted average	Concentration of Ambient Air		
		Industrial area	Residential rural & other area	Sensitive area
Sulphur dioxide (SO ₂)	Annual ^a	80 µg/m ³	60 µg/m ³	15 µg/m ³
	24 h ^b	120 µg/m ³	80 µg/m ³	30 µg/m ³
Oxides of nitrogen (NO ₂)	Annual ^a	80 µg/m ³	60 µg/m ³	15 µg/m ³
	24 h ^b	120 µg/m ³	80 µg/m ³	30 µg/m ³
Suspended Particulate Matter (SPM)	Annual ^a	360 µg/m ³	140 µg/m ³	70 µg/m ³
	24 h ^b	500 µg/m ³	200 µg/m ³	100 µg/m ³
Respirable Particulate Matter (size Less than 10 µm) RPM	Annual ^a	120 µg/m ³	60 µg/m ³	50 µg/m ³
	24 h ^b	150 µg/m ³	100 µg/m ³	75 µg/m ³
Lead as Pb	Annual ^a	1.0 µg/m ³	0.75 µg/m ³	0.50 µg/m ³
	24 h ^b	1.5 µg/m ³	1.0 µg/m ³	0.75 µg/m ³
Carbon monoxide	8 h ^b	5.0 mg/m ³	2.0 mg/m ³	1.0 mg/m ³
	1 h	10.0 mg/m ³	4.0 mg/m ³	2.0 mg/m ³

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